Unit 3 Review: Distributions — ANSWER KEY

3.1 Percentiles

1. 9.9 is at the **46.15**th percentile — 46 or 47 (rounded down or up) is fine

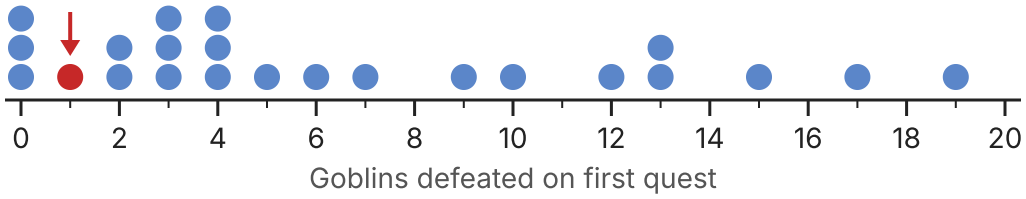
Values below 9.9: **5.3**, **6.1**, **8.0**, **8.0**, **8.7**, **9.7**, **[9.9]**, 10.7, 10.8, 11.3, 12.1, 12.3, 13.1 → **6** of the 13 values.
6 ÷ 13 ≈ 0.4615 → 46.15%
9.9 is at about the **46th percentile** (rounding down; 47th is fine too): about 46.2% of the data values are less than 9.9.

2. Score = **68** — answers that round to this are fine

We want 20% of 10 = 2 values below.
63, **65**, **[68]**, 81, 82, 82, 83, 83, 95, 97 → the value with exactly 2 values below it is **68**.
Check: **2 ÷ 10 = 20%**. So the 20th percentile is **68**.

3. Goblins = **1** goblins — answers that round to this are fine

We want 13% of 23 = 2.99 ≈ **3** values below.
Counting from the left: 3 at 0 → that is 3 dots, so the next value, **1**, has exactly 3 values below it.
Check: **3 ÷ 23 ≈ 13.04%** ≈ 13%. So the 13th percentile is **1**.
The 13th-percentile value, marked in red:



Goblins defeated on first quest

4. **C. 63% of 4-year-olds are shorter than Mia.**

The 63rd percentile means 63% of the values in the group are less than this one.
So the correct statement is: **63% of 4-year-olds are shorter than Mia.**

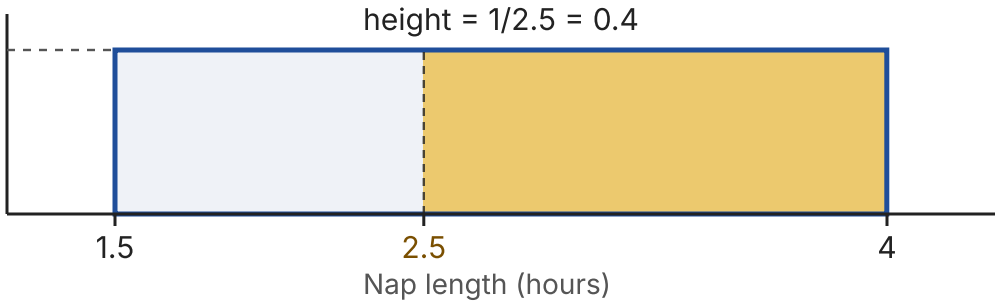
5. **C. It stays the same.**

Before: **20 ÷ 25 = 80%**. After: **20 ÷ 25 = 80%**.
The top score was already above, and it is still above. So the percentile **stays the same**.

3.2 Uniform Distributions

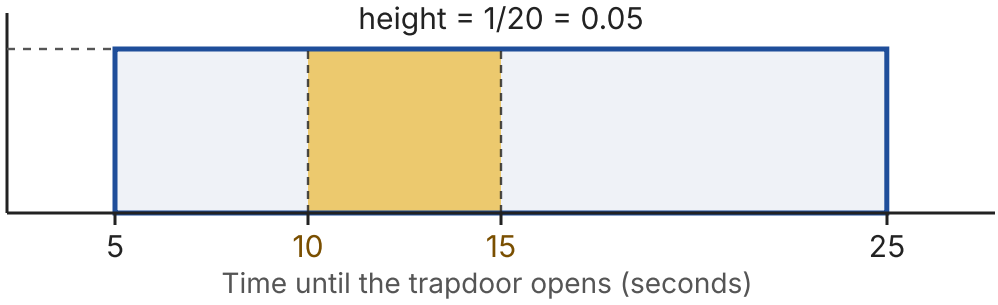
6. Percent = **60** % — answers that round to this are fine

Density curve: a rectangle from 1.5 to 4, height = **1 ÷ 2.5 = 1/2.5 = 0.4**.
Width of the shaded region: **4 − 2.5 = 1.5**
Area: **1.5 ÷ 2.5 = 0.6** → **60%**



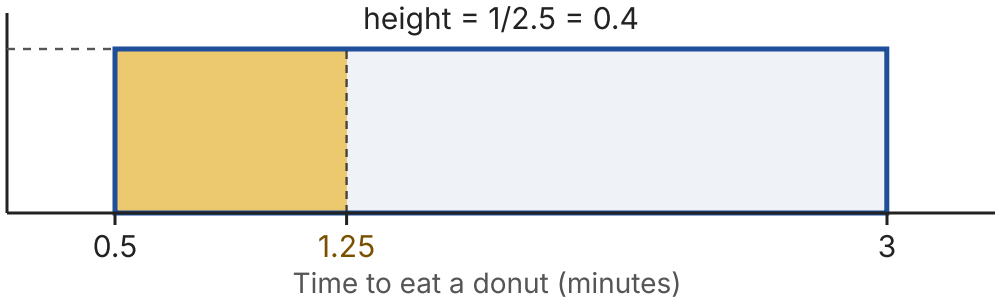
7. Percent = **25** % — answers that round to this are fine

Density curve: a rectangle from 5 to 25, height = **1 ÷ 20 = 1/20 = 0.05**.
Width of the shaded region: **15 − 10 = 5**
Area: **5 ÷ 20 = 0.25** → **25%**



8. 30th percentile = **1.25** minutes — answers that round to this are fine

Density curve: a rectangle from 0.5 to 3, height = **1 ÷ 2.5 = 1/2.5 = 0.4**.
We need 30% of the area to the left: width = **0.3 × 2.5 = 0.75**
Start at a = 0.5: **0.5 + 0.75 = 1.25**, so the 30th percentile is **1.25** minutes.



9. **A. From 30 to 60, height ≈ 0.03333**

Width = 30, so height = **1 ÷ 30 ≈ 0.0333** (then width × height = 1).
Correct: **From 30 to 60, height ≈ 0.03333**

10. **D. About 86.67% of the time, a driver will wait less than 78 seconds.**

0.8667 as a percent is 86.67%, and it describes how often a driver will wait less than 78 seconds.
Correct: **About 86.67% of the time, a driver will wait less than 78 seconds.**

3.3 z-Scores

11. z = **2.20** — answers that round to this are fine

z = (165 − 110) ÷ 25 = 55 ÷ 25 = 2.20
So the value is 2.2 standard deviations above the mean.

12. **A. Liam**

Liam: **z = (181 − 162.3) ÷ 27.2 = 18.7 ÷ 27.2 ≈ 0.69**
Chloe: **z = (224 − 246.7) ÷ 48.1 = −22.7 ÷ 48.1 ≈ −0.47**
Here higher is better, so the **larger** z-score wins. 0.69 vs. −0.47 → **Liam** did better relative to their group.

13. Weight = **9.215** oz — answers that round to this are fine

x = μ + z·σ = 9.12 + (1.9)(0.05)
= 9.12 + 0.095 = 9.215 → 9.215 ounces

14. **A. 48 hours below the mean**

0.6 standard deviations × 80 hours each = 48 hours.
Negative/positive z → below the mean. Correct: **48 hours below the mean**

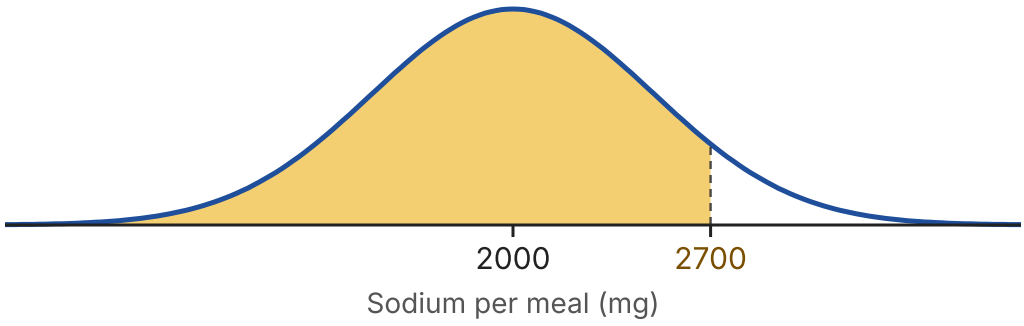
15. **D. 790**

430: z = (430 − 528) ÷ 117 ≈ −0.84
620: z = (620 − 528) ÷ 117 ≈ 0.79
490: z = (490 − 528) ÷ 117 ≈ −0.32
790: z = (790 − 528) ÷ 117 ≈ 2.24 ← **|z| > 2, unusual**

3.4 Normal Distributions (Desmos)

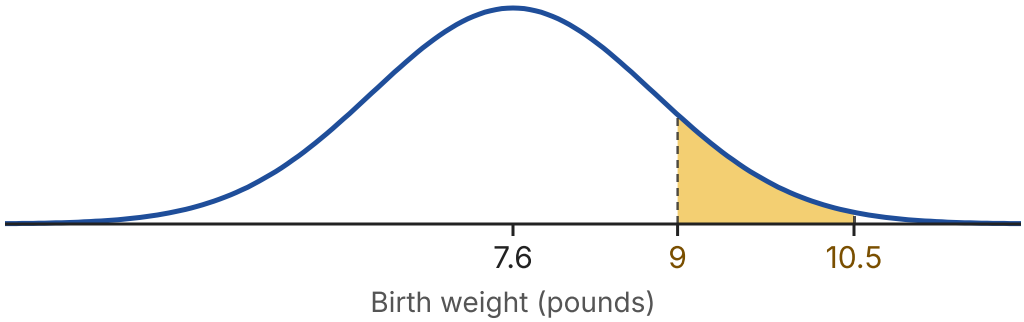
16. Percent = **91.92** % — 91 or 92 (rounded down or up) is fine

Desmos: **normaldist(2000, 500)**, cumulative probability from −∞ to 2700 → 0.9192
So about **91.92%** of meals have less than 2700 mg of sodium.
Sense check: z = (2700 − 2000) ÷ 500 ≈ 1.40, so 2700 mg is 1.4 standard deviations above the mean — the shaded area should be more than half.
Shaded region:



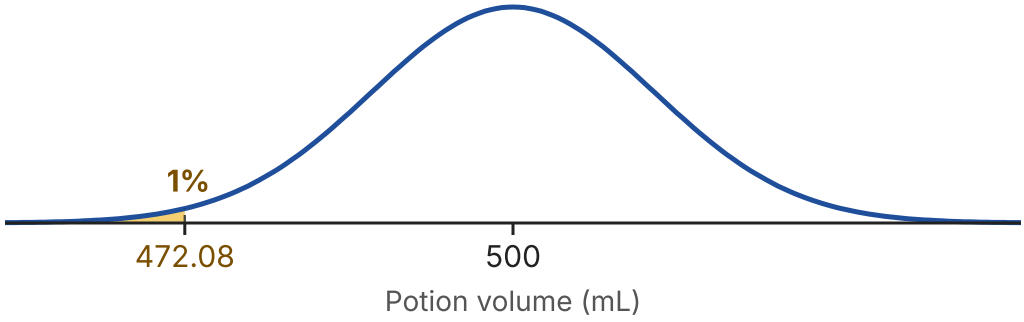
17. Percent = **11.38** % — 11 or 12 (rounded down or up) is fine

Desmos: **normaldist(7.6, 1.2)**, cumulative probability from 9 to 10.5 → 0.1138
So about **11.38%** of babies weigh between 9 and 10.5 pounds.
Shaded region:

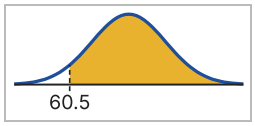


18. Cutoff = **472.08** mL — answers that round to this are fine

Bottom 1% → left area 0.01: **normaldist(500, 12).inversecdf(0.01)**
Cutoff ≈ **472.08 mL**
Shaded region:



19. **C**



"height is above 60.5 inches" means everything to the **right** of 60.5.

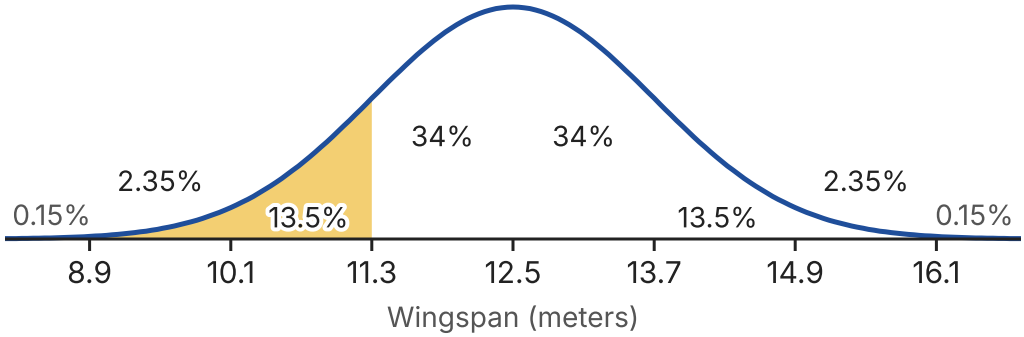
20. **B. About 15.87% of boxes weigh more than 16.5 ounces.**

0.1587 × 100 = 15.87%. The region from 16.5 to ∞ is above 16.5.
Correct: **About 15.87% of boxes weigh more than 16.5 ounces.**

3.5 The Empirical Rule

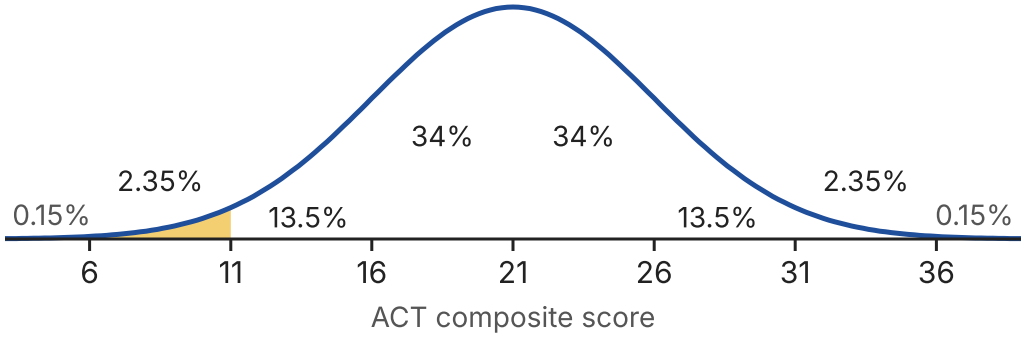
21. Proportion (as a percent) = **16** % — answers that round to this are fine

11.3 = 12.5 − 1.2, which is **1 SD below the mean**.
Area below it: **0.15 + 2.35 + 13.5** = **16%**
Shaded region:



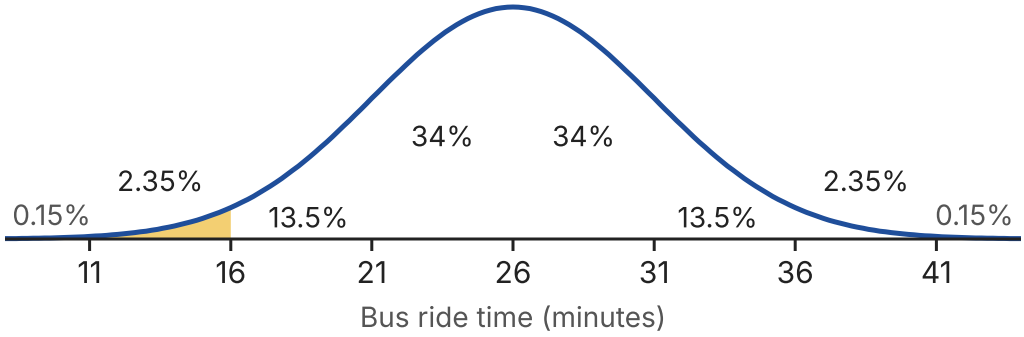
22. Percent = **2.35** % — 2 or 3 (rounded down or up) is fine

6 = 21 − 3 × 5, which is **3 SDs below the mean**. 11 = 21 − 2 × 5, which is **2 SDs below the mean**.
That is one piece of the 68–95–99.7 picture: **2.35%**
Shaded region:



23. Ride time = **16** minutes — answers that round to this are fine

Area to the left of μ − 2σ is **0.15 + 2.35 = 2.5%**.
So the 2.5th percentile is 2 SDs below the mean: **26 − 2 × 5** = **16 minutes**
Shaded region:

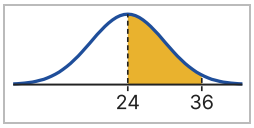


24. **D. 97.5%**

50% are below the mean, plus 47.5% between the mean and 2 standard deviations above: 50% + 47.5% = 97.5%.
So about **97.5%** of full charges have a battery life below the value that is 2 standard deviations above the mean.

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25. **C**



"fuel efficiency is between the mean and 2 standard deviations above the mean" means the middle part **between** 24 and 36.
